

Generating Synthetic Longitudinal Patient Data Using Stochastic Attention: Application to Coagulation Dynamics During Pregnancy

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Abstract

Small longitudinal clinical datasets pose a fundamental challenge for computational modeling and machine learning in biomedical research. We present a method for generating synthetic patient data using Stochastic Attention (SA), a sampling framework based on modern Hopfield networks. We apply SA to a longitudinal dataset of coagulation, fibrinolysis, thrombin generation, and viscoelastic measurements collected from pregnant patients across three clinical visits (pre-pregnancy baseline, first trimester, and third trimester). Our approach concatenates multi-visit patient profiles into unified patterns, applies PCA for dimensionality reduction, and uses Langevin dynamics on the Hopfield energy landscape to generate novel synthetic trajectories. A direction-magnitude decomposition preserves the natural variance structure of continuous clinical measurements. We demonstrate that SA-generated synthetic patients faithfully reproduce per-feature summary statistics (mean relative error < 3%), cross-feature correlation structure, and the characteristic pregnancy-driven shift in the Factor X versus Maximum Clot Firmness (MCF) plane. Compared with multivariate normal sampling, SA better preserves non-Gaussian marginal distributions and nonlinear inter-feature dependencies while generating clinically plausible novel patient trajectories.

Keywords: synthetic data generation, stochastic attention, modern Hopfield networks, coagulation, pregnancy, longitudinal data

1 Introduction

Pregnancy induces profound changes in the coagulation and fibrinolytic systems, shifting the hemostatic balance toward a hypercoagulable state [Grouzi et al., 2022]. Pregnant and postpartum women face a 4–5 fold increased risk of venous thromboembolism compared with non-pregnant women, and between 2017 and 2019, thrombotic embolisms caused 8.7% of pregnancy-related deaths in the United States, while hemorrhage caused 13.7% [Trost et al., 2022, Harris, 2023]. Understanding patient-specific coagulation dynamics during pregnancy is therefore critical for risk stratification and clinical decision-making.

However, longitudinal clinical datasets capturing the full coagulation, fibrinolysis, thrombin generation, and viscoelastic profiles across pregnancy are rare and expensive to collect. Typical cohort sizes are on the order of tens of patients, far too small for training machine learning models or performing robust statistical analyses. Synthetic data generation offers a path forward: by augmenting small real datasets with statistically faithful synthetic records, we can enable downstream computational analyses that would otherwise be infeasible.

In prior work, we generated synthetic patient populations using multivariate normal distributions fitted independently to each clinical visit [Bravo et al., 2023]. While this approach captured first- and second-order statistics, it assumed Gaussian marginals and linear correlations, and it generated each visit independently rather than as part of a coherent longitudinal trajectory.

Stochastic Attention (SA) provides an alternative framework rooted in modern Hopfield network theory [Ramsauer et al., 2021]. Modern Hopfield networks define an energy landscape over continuous state vectors:

$$E(\boldsymbol{\xi}) = \frac{1}{2} \|\boldsymbol{\xi}\|^2 - \frac{1}{\beta} \log \sum_{k=1}^K \exp \left(\beta \mathbf{m}_k^\top \boldsymbol{\xi} \right) \quad (1)$$

where $\{\mathbf{m}_k\}_{k=1}^K$ are stored memory patterns and β is an inverse temperature parameter controlling the sharpness of retrieval. By running Langevin dynamics on this energy landscape, SA generates novel samples that interpolate between stored patterns in a manner governed by the attention mechanism.

In this work, we adapt SA for continuous clinical data and longitudinal patient trajectories. Our contributions are:

1. We formulate a pipeline for applying SA to continuous multivariate clinical measurements, including a direction-magnitude decomposition that preserves anisotropic variance structure lost by unit-norm projection.
2. We introduce concatenated-visit patterns that treat each patient’s full longitudinal profile as an atomic unit, enabling SA to generate coherent multi-visit trajectories.
3. We demonstrate that SA-generated synthetic patients faithfully reproduce summary statistics, correlation structure, and biologically meaningful pregnancy trajectories from a 72-feature, 3-visit coagulation dataset.
4. We compare SA against multivariate normal generation, showing improved preservation of non-Gaussian features and nonlinear dependencies.

2 Methods

2.1 Clinical Dataset

We analyzed a longitudinal dataset of coagulation, fibrinolysis, thrombin generation assay (TGA), and rotational thromboelastometry (ROTEM) measurements collected from pregnant patients at three clinical visits: Visit 1 (pre-pregnancy baseline), Visit 2 (first trimester), and Visit 3 (third trimester). The dataset contained 153 total records across approximately 50 subjects. After removing columns with $> 30\%$ missing values and retaining only subjects with complete data across all three visits, we obtained $K = 23$ complete longitudinal profiles across $n = 72$ assay measurements per visit.

The 72 assay measurements spanned five categories: (i) hormones (estradiol, progesterone), (ii) coagulation factors and inhibitors (Factors II, V, VII, VIII, IX, X, XI, XII; AT, PC, TFPI, vWF, fibrinogen, etc.), (iii) fibrinolytic markers (plasminogen, PAI-1, PAI-2, α_2 -AP, TAFI, tPA, D-dimer equivalents), (iv) thrombin generation parameters under four initiator conditions (TF, TF+TM, No TF, No TF+anti-FXIa), and (v) ROTEM/viscoelastic parameters (CT, MCF, alpha angle, LOT, ML, LT, AUC under TF Only and TF+tPA conditions).

2.2 Stochastic Attention Framework

2.2.1 Modern Hopfield Energy and Langevin Sampling

We store K memory patterns as columns of a matrix $\hat{\mathbf{X}} \in \mathbb{R}^{d \times K}$ and sample from the associated Hopfield energy landscape (Eq. 1) using an Unadjusted Langevin Algorithm (ULA):

$$\boldsymbol{\xi}_{t+1} = (1 - \alpha) \boldsymbol{\xi}_t + \alpha \hat{\mathbf{X}} \text{softmax}\left(\beta \hat{\mathbf{X}}^\top \boldsymbol{\xi}_t\right) + \sqrt{\frac{2\alpha}{\beta}} \boldsymbol{\varepsilon}_t \quad (2)$$

where α is a step size, β is the inverse temperature, and $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The deterministic component drives $\boldsymbol{\xi}_t$ toward an attention-weighted combination of memory patterns, while the stochastic component enables exploration and novelty.

2.2.2 Phase Transition and β^* Selection

The inverse temperature β controls the trade-off between retrieval (large β , converging to nearest memory) and generation (small β , uniform mixing). We identify the critical β^* at the phase transition by computing the normalized attention entropy:

$$\bar{H}(\beta) = -\frac{1}{\log K} \sum_{k=1}^K p_k \log p_k, \quad p_k = \text{softmax}\left(\beta \hat{\mathbf{X}}^\top \boldsymbol{\xi}\right)_k \quad (3)$$

and finding the inflection point (maximum of the numerical second derivative) of $\bar{H}(\beta)$ over a logarithmic sweep of β values. Operating near β^* produces samples that are novel yet statistically consistent with the stored patterns.

2.3 Pipeline for Continuous Clinical Data

2.3.1 Longitudinal Concatenation

For each of the $K = 23$ subjects with complete data, we concatenate the $n = 72$ assay measurements across all three visits into a single vector of dimension $d_{\text{concat}} = 3n = 216$:

$$\mathbf{x}_k = \begin{bmatrix} \mathbf{x}_k^{(V1)} \\ \mathbf{x}_k^{(V2)} \\ \mathbf{x}_k^{(V3)} \end{bmatrix} \in \mathbb{R}^{216} \quad (4)$$

This ensures that generated synthetic patients have internally consistent longitudinal trajectories.

2.3.2 Standardization and PCA

We standardize each feature to zero mean and unit variance, then apply PCA retaining 95% of the variance, reducing dimensionality from $d_{\text{concat}} = 216$ to $d_{\text{PCA}} = 18$. This yields a memory-to-dimension ratio of $K/d_{\text{PCA}} \approx 1.28$, which is in the regime where SA generates diverse yet faithful samples.

2.3.3 Direction-Magnitude Decomposition (Rescaled SA)

Standard SA operates on unit-norm patterns: $\hat{\mathbf{m}}_k = \mathbf{m}_k / \|\mathbf{m}_k\|$. While appropriate for discrete data (e.g., one-hot protein sequences), unit-norm projection destroys the anisotropic variance structure of continuous clinical data, collapsing dispersion in all principal components.

We address this with a direction-magnitude decomposition:

1. Record the PCA-space norms $\{r_k = \|\mathbf{m}_k\|\}_{k=1}^K$ before normalization.
2. Run SA on unit-norm patterns to obtain a direction $\hat{\xi}$.
3. Draw a magnitude r from the empirical distribution of $\{r_k\}$.
4. Reconstruct: $\xi_{\text{rescaled}} = r \cdot \hat{\xi}$.

This preserves the directional structure learned by SA while restoring the natural scale of variation.

2.3.4 Inverse Transform

The rescaled PCA vector is mapped back to the original feature space via inverse PCA and de-standardization:

$$\mathbf{x}_{\text{synth}} = \boldsymbol{\sigma} \odot \left(\mathbf{W}^\top \xi_{\text{rescaled}} + \bar{\mathbf{z}} \right) + \boldsymbol{\mu} \quad (5)$$

where $\mathbf{W}$ is the PCA loading matrix, $\bar{\mathbf{z}}$ is the PCA mean, and $\boldsymbol{\mu}, \boldsymbol{\sigma}$ are the original feature means and standard deviations. The 216-dimensional synthetic vector is then split into three 72-dimensional visit records.

3 Results

3.1 Summary Statistics

3.2 PCA Projection by Visit

3.3 Correlation Structure Preservation

3.4 Factor X vs MCF Trajectory

3.5 Comparison with Multivariate Normal Generation

3.6 Novelty Analysis

3.7 Sample Size Sensitivity

4 Discussion

5 Conclusion

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